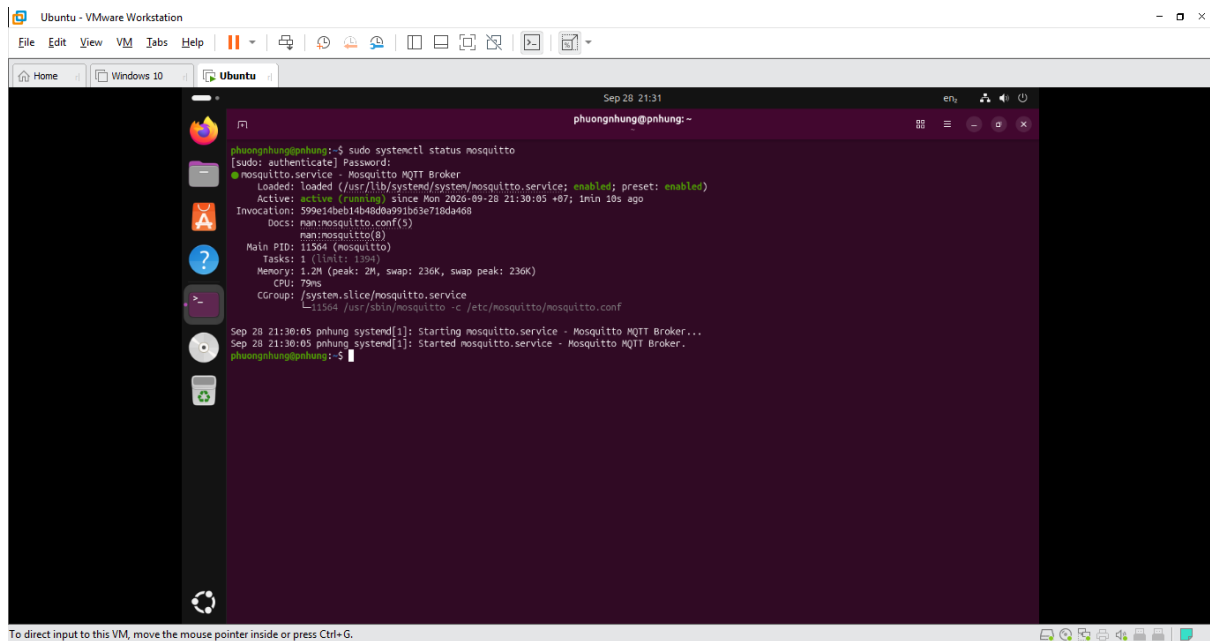


BÀI THỰC HÀNH TRÊN LỚP

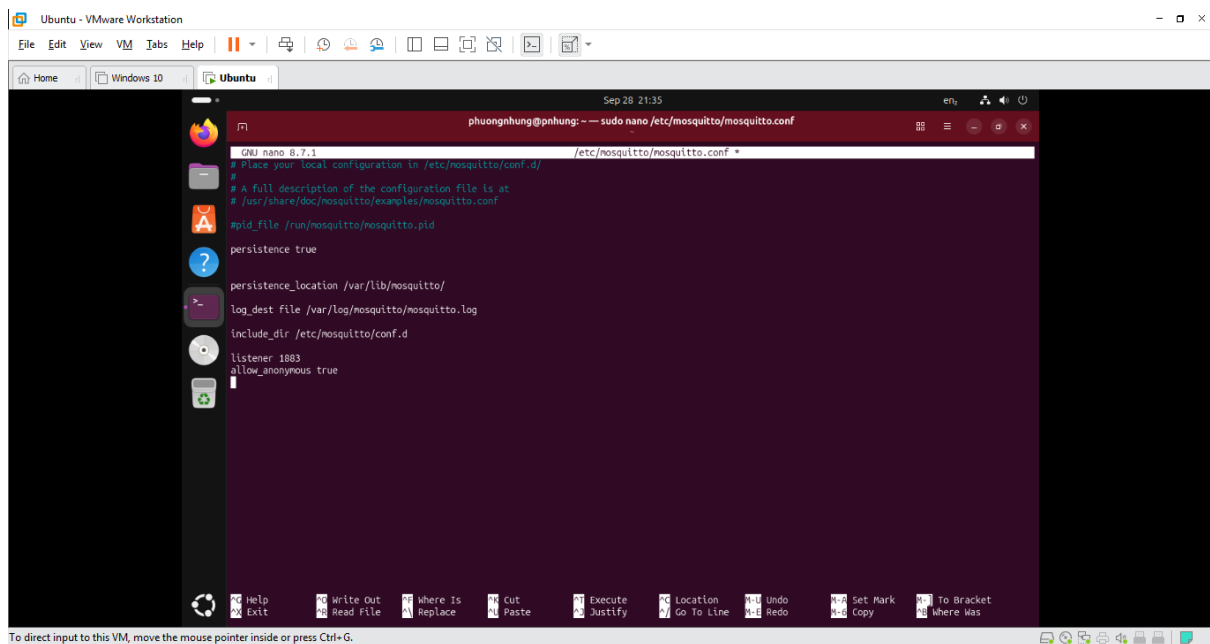
• Bài 1: Cài đặt và cấu hình Mosquitto Local Broker trên Gateway



The screenshot shows a terminal window in a VMware Workstation environment. The user is logged in as 'phuongnhung' on an Ubuntu system. The terminal output shows the command 'sudo systemctl status mosquitto' being executed, which returns the status of the 'mosquitto.service'. The service is listed as 'loaded' and 'enabled'. The active state is 'active (running)' since Mon 2026-09-28 21:30:05 +07; 1min 10s ago. The invocation is '599e140eb14b460ba991d63e718da408'. The docs are 'man:mosquitto.conf(8)'. The main PID is 11564 (mosquitto). The tasks are 1 (limit: 1394). The memory is 1.2M (peak: 2M, swap: 236K, swap peak: 236K). The CPU is 79ms. The CGroup is '/system.slice/mosquitto.service'. The command 'systemctl start mosquitto.service' is also shown, and the service is started.

```
phuongnhung@phung:~$ sudo systemctl status mosquitto
[sudo: authenticate] Password:
● mosquitto.service - Mosquitto MQTT Broker
   Loaded: loaded (/usr/lib/systemd/system/mosquitto.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-09-28 21:30:05 +07; 1min 10s ago
     Invocation: 599e140eb14b460ba991d63e718da408
       Docs: man:mosquitto.conf(8)
   Main PID: 11564 (mosquitto)
    Tasks: 1 (limit: 1394)
  Memory: 1.2M (peak: 2M, swap: 236K, swap peak: 236K)
     CPU: 79ms
    CGroup: /system.slice/mosquitto.service
            └─11564 /usr/sbin/mosquitto -c /etc/mosquitto/mosquitto.conf

Sep 28 21:30:05 phung systemd[1]: Starting mosquitto.service - Mosquitto MQTT Broker...
Sep 28 21:30:05 phung systemd[1]: Started mosquitto.service - Mosquitto MQTT Broker.
phuongnhung@phung:~$
```



The screenshot shows a terminal window in a VMware Workstation environment. The user is logged in as 'phuongnhung' on an Ubuntu system. The terminal output shows the command 'sudo nano /etc/mosquitto/mosquitto.conf' being executed, which opens the configuration file in the nano editor. The configuration file contains the following settings:

```
GNU nano 8.7.1 /etc/mosquitto/mosquitto.conf
# Place your local configuration in /etc/mosquitto/conf.d/
#
# A full description of the configuration file is at
# /usr/share/doc/mosquitto/examples/mosquitto.conf
#pid_file /run/mosquitto/mosquitto.pid

persistence true

persistence_location /var/lib/mosquitto/

log_dest file /var/log/mosquitto/mosquitto.log

include_dir /etc/mosquitto/conf.d

listener 1883
  allow_anonymous true
```

```
phuongnhung@phung:~$ sudo systemctl status mosquitto
[sudo: authenticate] Password:
● mosquitto.service - Mosquitto MQTT Broker
   Loaded: loaded (/usr/lib/systemd/system/mosquitto.service; enabled; preset: enabled)
   Active: active (running) since Mon 2020-09-28 21:38:05 +07; 18s ago
     Invocation: 599e14beb14b48da991b63e718da408
       Docs: man:mosquitto.conf(5)
             man:mosquitto(8)
    Main PID: 11504 (mosquitto)
      Tasks: 1 (limit: 1394)
     Memory: 1.2M (peak: 2M, swap: 236K, swap peak: 236K)
        CPU: 79ms
     CGroup: /system.slice/mosquitto.service
            └─11504 /usr/sbin/mosquitto -c /etc/mosquitto/mosquitto.conf

Sep 28 21:38:05 phung systemd[1]: Starting mosquitto.service - Mosquitto MQTT Broker...
Sep 28 21:38:05 phung systemd[1]: Started mosquitto.service - Mosquitto MQTT Broker.
phuongnhung@phung:~$ sudo nano /etc/mosquitto/mosquitto.conf
phuongnhung@phung:~$ sudo systemctl restart mosquitto
phuongnhung@phung:~$ sudo systemctl status mosquitto
● mosquitto.service - Mosquitto MQTT Broker
   Loaded: loaded (/usr/lib/systemd/system/mosquitto.service; enabled; preset: enabled)
   Active: active (running) since Mon 2020-09-28 21:38:40 +07; 7s ago
     Invocation: 5fa9fcee74c440b8d0d06e01e169638
       Docs: man:mosquitto.conf(5)
             man:mosquitto(8)
    Main PID: 16260 (mosquitto)
      Tasks: 1 (limit: 1394)
     Memory: 1.1M (peak: 1.7M)
        CPU: 26ms
     CGroup: /system.slice/mosquitto.service
            └─16260 /usr/sbin/mosquitto -c /etc/mosquitto/mosquitto.conf

Sep 28 21:38:40 phung systemd[1]: Starting mosquitto.service - Mosquitto MQTT Broker...
Sep 28 21:38:40 phung systemd[1]: Started mosquitto.service - Mosquitto MQTT Broker.
phuongnhung@phung:~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

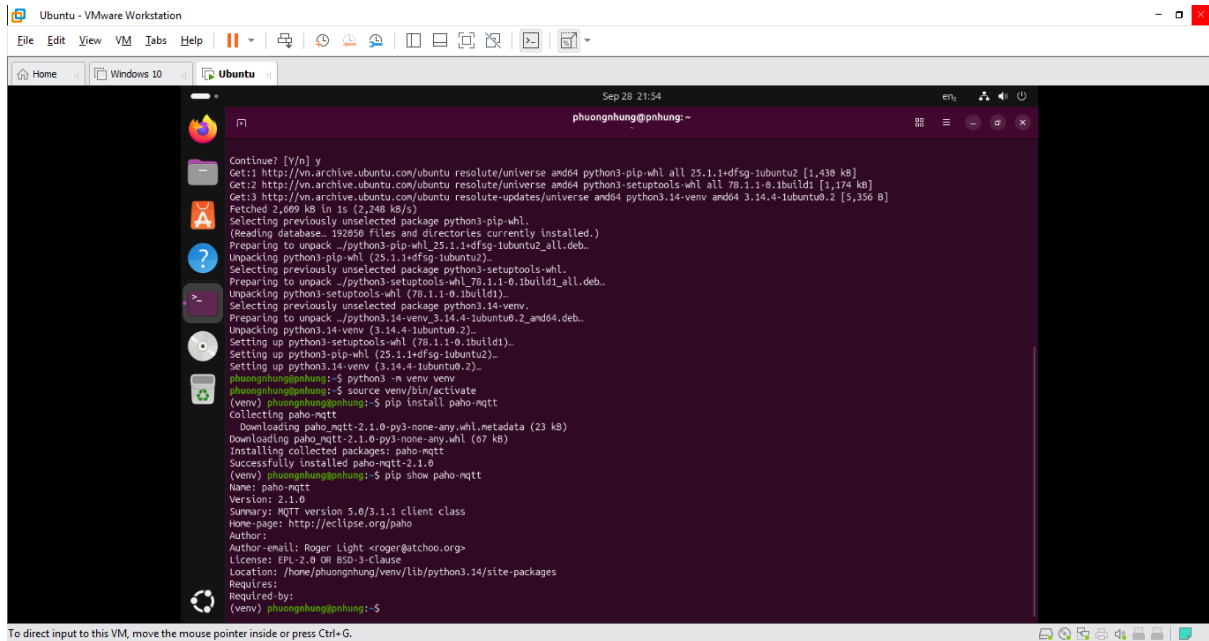
• Bài 2: Kiểm tra kết nối MQTT Client bằng dòng lệnh (CLI)

```
phuongnhung@phung:~$ mosquitto_sub -h localhost -t gateway/test
Hello Local MQTT

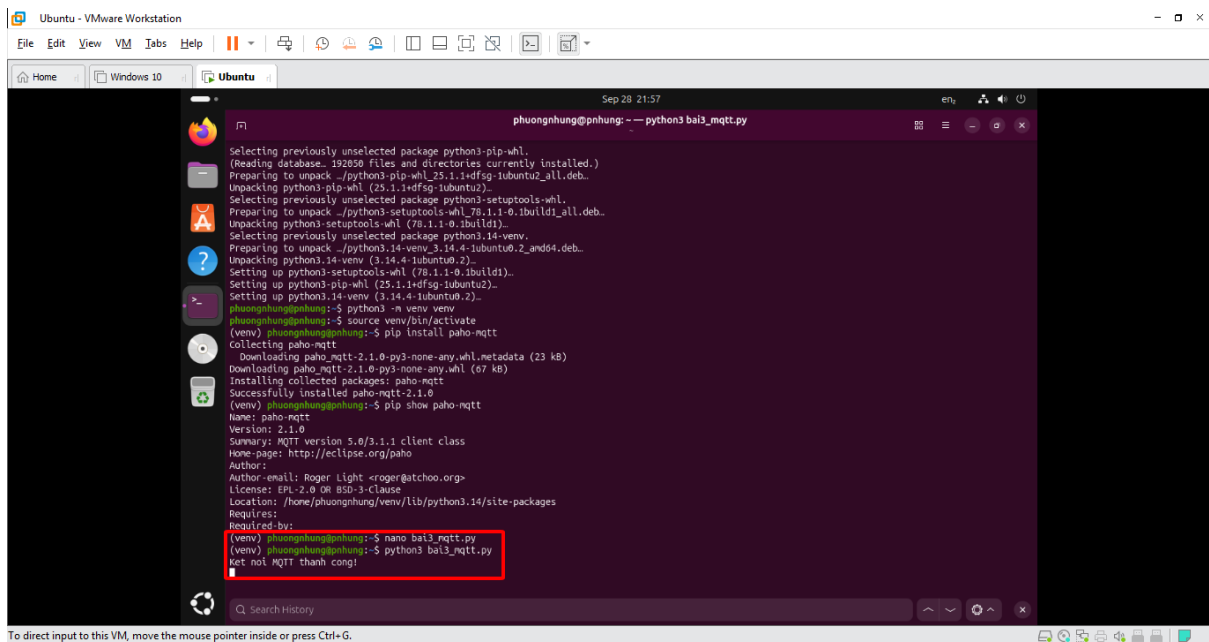
phuongnhung@phung:~$ mosquitto_pub -h localhost -t "gateway/test" -m "Hello Local MQTT"
phuongnhung@phung:~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

• Bài 3: Cài đặt thư viện Python Paho-MQTT trên Gateway

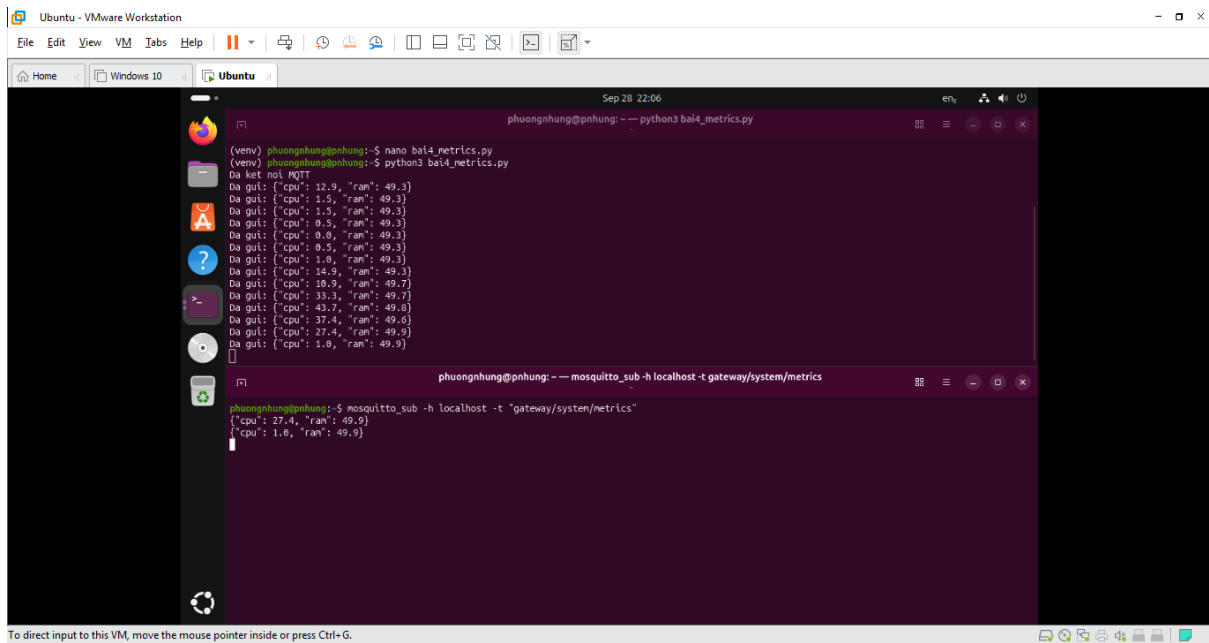


```
phuongnhung@phung: ~$ sudo apt-get install python3-pip python3-setuptools python3-venv
Get:1 http://vn.archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pip-whl all 25.1.1+dfsg-1ubuntu2 [1,430 kB]
Get:2 http://vn.archive.ubuntu.com/ubuntu resolute/universe amd64 python3-setuptools-whl all 78.1.1-0.1build1 [1,174 kB]
Get:3 http://vn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 python3.14-venv amd64 3.14.4-1ubuntu0.2 [5,356 B]
Fetched 2,609 kB in 1s (2,240 kB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database... 192050 files and directories currently installed.)
Preparing to unpack ./python3-pip-whl_25.1.1+dfsg-1ubuntu2_all.deb...
Unpacking python3-pip-whl (25.1.1+dfsg-1ubuntu2)...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack ./python3-setuptools-whl_78.1.1-0.1build1_all.deb...
Unpacking python3-setuptools-whl (78.1.1-0.1build1)...
Selecting previously unselected package python3.14-venv.
Preparing to unpack ./python3.14-venv_3.14.4-1ubuntu0.2_and64.deb...
Unpacking python3.14-venv (3.14.4-1ubuntu0.2)...
Setting up python3-setuptools-whl (78.1.1-0.1build1)...
Setting up python3-pip-whl (25.1.1+dfsg-1ubuntu2)...
Setting up python3.14-venv (3.14.4-1ubuntu0.2)...
phuongnhung@phung: ~$ python3 -m venv venv
phuongnhung@phung: ~$ source venv/bin/activate
(venv) phuongnhung@phung: ~$ pip install paho-mqtt
Collecting paho-mqtt
  Downloading paho-mqtt-2.1.0-py3-none-any.whl.metadata (23 kB)
  Downloading paho-mqtt-2.1.0-py3-none-any.whl (67 kB)
Installing collected packages: paho-mqtt
Successfully installed paho-mqtt-2.1.0
(venv) phuongnhung@phung: ~$ pip show paho-mqtt
Name: paho-mqtt
Version: 2.1.0
Summary: MQTT version 5.0/3.1.1 client class
Home-page: http://eclipse.org/paho
Author:
Author-email: Roger Light <roger@atchoo.org>
License: EPL-2.0 OR BSD-3-Clause
Location: /home/phuongnhung/venv/lib/python3.14/site-packages
Requires:
Required-by:
(venv) phuongnhung@phung: ~$
```



```
phuongnhung@phung: ~$ sudo apt-get install python3-pip python3-setuptools python3-venv
Get:1 http://vn.archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pip-whl all 25.1.1+dfsg-1ubuntu2 [1,430 kB]
Get:2 http://vn.archive.ubuntu.com/ubuntu resolute/universe amd64 python3-setuptools-whl all 78.1.1-0.1build1 [1,174 kB]
Get:3 http://vn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 python3.14-venv amd64 3.14.4-1ubuntu0.2 [5,356 B]
Fetched 2,609 kB in 1s (2,240 kB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database... 192050 files and directories currently installed.)
Preparing to unpack ./python3-pip-whl_25.1.1+dfsg-1ubuntu2_all.deb...
Unpacking python3-pip-whl (25.1.1+dfsg-1ubuntu2)...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack ./python3-setuptools-whl_78.1.1-0.1build1_all.deb...
Unpacking python3-setuptools-whl (78.1.1-0.1build1)...
Selecting previously unselected package python3.14-venv.
Preparing to unpack ./python3.14-venv_3.14.4-1ubuntu0.2_and64.deb...
Unpacking python3.14-venv (3.14.4-1ubuntu0.2)...
Setting up python3-setuptools-whl (78.1.1-0.1build1)...
Setting up python3-pip-whl (25.1.1+dfsg-1ubuntu2)...
Setting up python3.14-venv (3.14.4-1ubuntu0.2)...
phuongnhung@phung: ~$ python3 -m venv venv
phuongnhung@phung: ~$ source venv/bin/activate
(venv) phuongnhung@phung: ~$ pip install paho-mqtt
Collecting paho-mqtt
  Downloading paho-mqtt-2.1.0-py3-none-any.whl.metadata (23 kB)
  Downloading paho-mqtt-2.1.0-py3-none-any.whl (67 kB)
Installing collected packages: paho-mqtt
Successfully installed paho-mqtt-2.1.0
(venv) phuongnhung@phung: ~$ pip show paho-mqtt
Name: paho-mqtt
Version: 2.1.0
Summary: MQTT version 5.0/3.1.1 client class
Home-page: http://eclipse.org/paho
Author:
Author-email: Roger Light <roger@atchoo.org>
License: EPL-2.0 OR BSD-3-Clause
Location: /home/phuongnhung/venv/lib/python3.14/site-packages
Requires:
Required-by:
(venv) phuongnhung@phung: ~$ nano b013_mqtt.py
(venv) phuongnhung@phung: ~$ python3 b013_mqtt.py
Ket noi MQTT thanh cong!
```

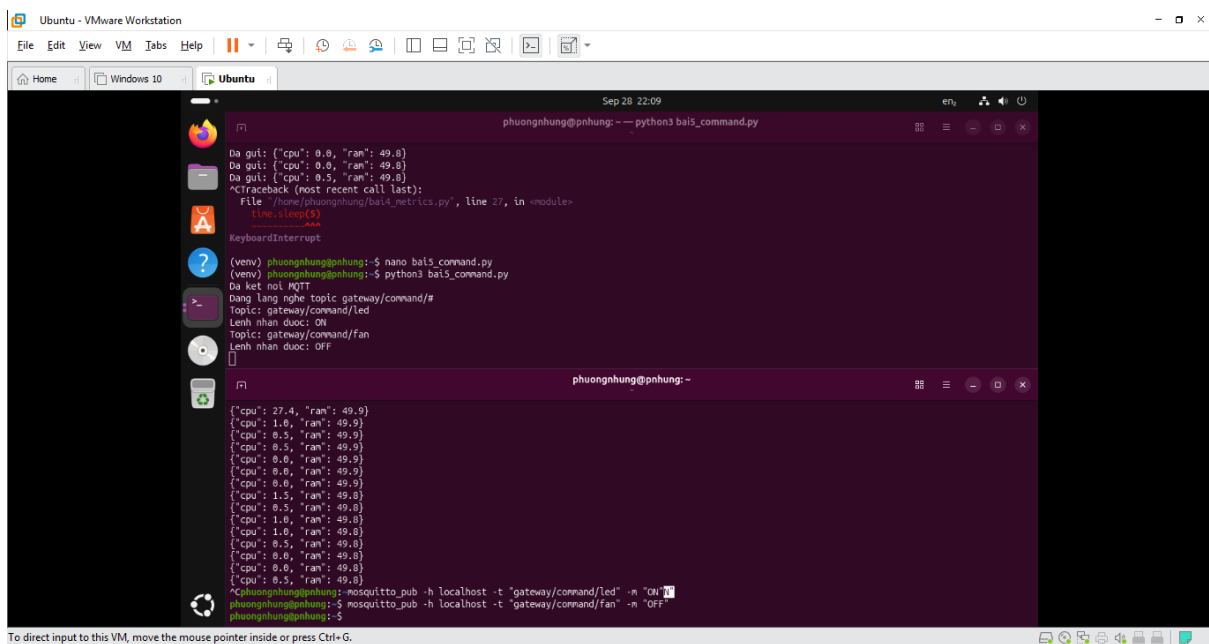
- **Bài 4: Lập trình Gateway Publish dữ liệu phần cứng hệ thống định kỳ**



```
phuongnhung@pnhung: ~ -- python3 ba4_metrics.py
(venv) phuongnhung@pnhung:~$ nano ba4_metrics.py
(venv) phuongnhung@pnhung:~$ python3 ba4_metrics.py
Đã kết nối MQTT
Đã gửi: [{"cpu": 12.9, "ram": 49.3}]
Đã gửi: [{"cpu": 1.5, "ram": 49.3}]
Đã gửi: [{"cpu": 1.5, "ram": 49.3}]
Đã gửi: [{"cpu": 0.5, "ram": 49.3}]
Đã gửi: [{"cpu": 0.0, "ram": 49.3}]
Đã gửi: [{"cpu": 0.5, "ram": 49.3}]
Đã gửi: [{"cpu": 1.0, "ram": 49.3}]
Đã gửi: [{"cpu": 14.9, "ram": 49.3}]
Đã gửi: [{"cpu": 10.9, "ram": 49.7}]
Đã gửi: [{"cpu": 33.3, "ram": 49.7}]
Đã gửi: [{"cpu": 43.7, "ram": 49.8}]
Đã gửi: [{"cpu": 37.4, "ram": 49.6}]
Đã gửi: [{"cpu": 27.4, "ram": 49.9}]
Đã gửi: [{"cpu": 1.0, "ram": 49.9}]

phuongnhung@pnhung: ~ -- mosquitto_sub -h localhost -t gateway/system/metrics
phuongnhung@pnhung:~$ mosquitto_sub -h localhost -t "gateway/system/metrics"
{"cpu": 27.4, "ram": 49.9}
{"cpu": 1.0, "ram": 49.9}
```

- **Bài 5: Lập trình Gateway Subscribe lắng nghe lệnh điều khiển nội bộ**



```
phuongnhung@pnhung: ~ -- python3 ba5_command.py
Đã gửi: [{"cpu": 0.0, "ram": 49.8}]
Đã gửi: [{"cpu": 0.0, "ram": 49.8}]
Đã gửi: [{"cpu": 0.5, "ram": 49.8}]
^CTraceback (most recent call last):
  File "/home/phuongnhung/ba4_metrics.py", line 27, in <module>
    time.sleep(5)
KeyboardInterrupt

(venv) phuongnhung@pnhung:~$ nano ba5_command.py
(venv) phuongnhung@pnhung:~$ python3 ba5_command.py
Đã kết nối MQTT
Đăng ký nghe topic gateway/command/#
Topic: gateway/command/led
Lệnh nhận được: ON
Topic: gateway/command/fan
Lệnh nhận được: OFF

phuongnhung@pnhung: ~
{"cpu": 27.4, "ram": 49.9}
{"cpu": 1.0, "ram": 49.0}
{"cpu": 0.5, "ram": 49.9}
{"cpu": 0.5, "ram": 49.9}
{"cpu": 0.0, "ram": 49.9}
{"cpu": 0.0, "ram": 49.8}
{"cpu": 0.0, "ram": 49.9}
{"cpu": 1.5, "ram": 49.8}
{"cpu": 0.5, "ram": 49.8}
{"cpu": 1.0, "ram": 49.8}
{"cpu": 0.5, "ram": 49.8}
{"cpu": 0.0, "ram": 49.8}
{"cpu": 0.5, "ram": 49.8}
{"cpu": 0.0, "ram": 49.8}
{"cpu": 0.5, "ram": 49.8}

phuongnhung@pnhung:~$ mosquitto_pub -h localhost -t "gateway/command/led" -m "ON"
phuongnhung@pnhung:~$ mosquitto_pub -h localhost -t "gateway/command/fan" -m "OFF"
phuongnhung@pnhung:~$
```

BÀI THỰC HÀNH VỀ NHÀ

- **Bài 1 (Về nhà): Gateway làm trạm tập trung dữ liệu (Data Aggregator)**

```
phuongnhung@phnhung: ~ -- python3 btvn1_aggregator.py
(venv) phuongnhung@phnhung:~$ python3 btvn1_aggregator.py
Đã kết nối MQTT
Đang lắng nghe topic sensor-/data
Đã gửi báo cáo: {}
Nhận từ sensor1: {'temp': 25, 'hum': 60}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}}
Nhận từ sensor2: {'temp': 30, 'hum': 55}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}
Đã gửi báo cáo: {'sensor1': {'temp': 25, 'hum': 60}, 'sensor2': {'temp': 30, 'hum': 55}}

phuongnhung@phnhung:~$ mosquitto_pub -h localhost -t "sensor/sensor1/data" -m '{"temp":25,"hum":60}'
phuongnhung@phnhung:~$ mosquitto_pub -h localhost -t "sensor/sensor2/data" -m '{"temp":30,"hum":55}'
phuongnhung@phnhung:~$

phuongnhung@phnhung:~ -- mosquitto_sub -h localhost -t gateway/aggregated_report

phuongnhung@phnhung:~$ mosquitto_sub -h localhost -t "gateway/aggregated_report"
{"sensor1": {"temp": 25, "hum": 60}, "sensor2": {"temp": 30, "hum": 55}}
{"sensor1": {"temp": 25, "hum": 60}, "sensor2": {"temp": 30, "hum": 55}}
```

- **Bài 2 (Về nhà): Cấu hình bảo mật xác thực cho Local Broker (Access Control)**

```
phuongnhung@phnhung: ~$ sudo mosquitto -c /etc/mosquitto/mosquitto.conf -v
phuongnhung@phnhung:~$ sudo chown mosquitto:mosquitto /etc/mosquitto/passwd
phuongnhung@phnhung:~$ sudo chmod 640 /etc/mosquitto/passwd
phuongnhung@phnhung:~$ sudo systemctl restart mosquitto
phuongnhung@phnhung:~$ sudo systemctl status mosquitto
● mosquitto.service - Mosquitto MQTT Broker
   Loaded: loaded (/usr/lib/systemd/system/mosquitto.service; enabled; preset: enabled)
   Active: active (running) since Mon 2020-09-28 22:22:36 +07; 9s ago
     Invocation: 42e7ceab6d594355de5507362796a2
       Docs: man:mosquitto.conf(5)
            man:mosquitto(8)
   Main PID: 18088 (mosquitto)
     Tasks: 1 (limit: 3394)
    Memory: 1.1M (peak: 2M)
       CPU: 45ms
   CGroup: /system.slice/mosquitto.service
           └─18088 /usr/sbin/mosquitto -c /etc/mosquitto/mosquitto.conf

Sep 28 22:22:36 phnhung systemd[j]: Starting mosquitto.service - Mosquitto MQTT Broker...
Sep 28 22:22:36 phnhung systemd[j]: Started mosquitto.service - Mosquitto MQTT Broker.
phuongnhung@phnhung:~$
```

Ubuntu - VMware Workstation

File Edit View VM Tabs Help

Home Ubuntu

Sep 28 22:24

phuongnhung@phung: ~

```
Subject: Unit failed
Defined-By: systemd
Support: http://www.ubuntu.com/support

The unit mosquitto.service has entered the 'failed' state with result 'exit-code'.
Sep 28 22:20:10 phung systemd[1]: Failed to start mosquitto.service - Mosquitto MQTT Broker.
Subject: A start job for unit mosquitto.service has failed
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit mosquitto.service has finished with a failure.

The job identifier is 5518 and the job result is failed.
phuongnhung@phung:~$ sudo mosquitto -c /etc/mosquitto/mosquitto.conf -v
phuongnhung@phung:~$ ^C
phuongnhung@phung:~$ sudo chown mosquitto:mosquitto /etc/mosquitto/passwd
phuongnhung@phung:~$ sudo chown 640 /etc/mosquitto/passwd
phuongnhung@phung:~$ sudo systemctl restart mosquitto
phuongnhung@phung:~$ sudo systemctl status mosquitto
● mosquitto.service - Mosquitto MQTT Broker
   Loaded: loaded (/usr/lib/systemd/system/mosquitto.service; enabled; preset: enabled)
   Active: active (running) since Mon 2020-09-28 22:22:36 +07; 9s ago
     Invocation: 42e7ceab86594355adee058736279b82
       Docs: man:mosquitto.conf(5)
    Main PID: 18088 (mosquitto)
      Tasks: 1 (limit: 1394)
     Memory: 1.1M (peak: 2M)
        CPU: 45ms
    CGroup: /system.slice/mosquitto.service
            └─18088 /usr/sbin/mosquitto -c /etc/mosquitto/mosquitto.conf

Sep 28 22:22:36 phung systemd[1]: Starting mosquitto.service - Mosquitto MQTT Broker...
Sep 28 22:22:36 phung systemd[1]: Started mosquitto.service - Mosquitto MQTT Broker.
phuongnhung@phung:~$ mosquitto_sub -h localhost -t "test"
Connection error: Connection Refused: not authorised.
phuongnhung@phung:~$ mosquitto_sub -h localhost -t "test"
Connection error: Connection Refused: not authorised.
phuongnhung@phung:~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl-G.

Ubuntu - VMware Workstation

File Edit View VM Tabs Help

Home Ubuntu

Sep 28 22:25

phuongnhung@phung: ~

```
Docs: man:mosquitto.conf(5)
man:mosquitto(8)
Main PID: 18088 (mosquitto)
Tasks: 1 (limit: 1394)
Memory: 1.1M (peak: 2M)
CPU: 45ms
CGroup: /system.slice/mosquitto.service
        └─18088 /usr/sbin/mosquitto -c /etc/mosquitto/mosquitto.conf

Sep 28 22:22:36 phung systemd[1]: Starting mosquitto.service - Mosquitto MQTT Broker...
Sep 28 22:22:36 phung systemd[1]: Started mosquitto.service - Mosquitto MQTT Broker.
phuongnhung@phung:~$ mosquitto_sub -h localhost -t "test"
Connection error: Connection Refused: not authorised.
phuongnhung@phung:~$ mosquitto_sub -h localhost -t "test"
Connection error: Connection Refused: not authorised.
phuongnhung@phung:~$ mosquitto_sub -h localhost -t "test" -u iotuser -P 123456
Hello Secure MQTT

phuongnhung@phung:~$ mosquitto_pub -h localhost -t "test" -m "Hello Secure MQTT" -u iotuser -P 123456
phuongnhung@phung:~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl-G.

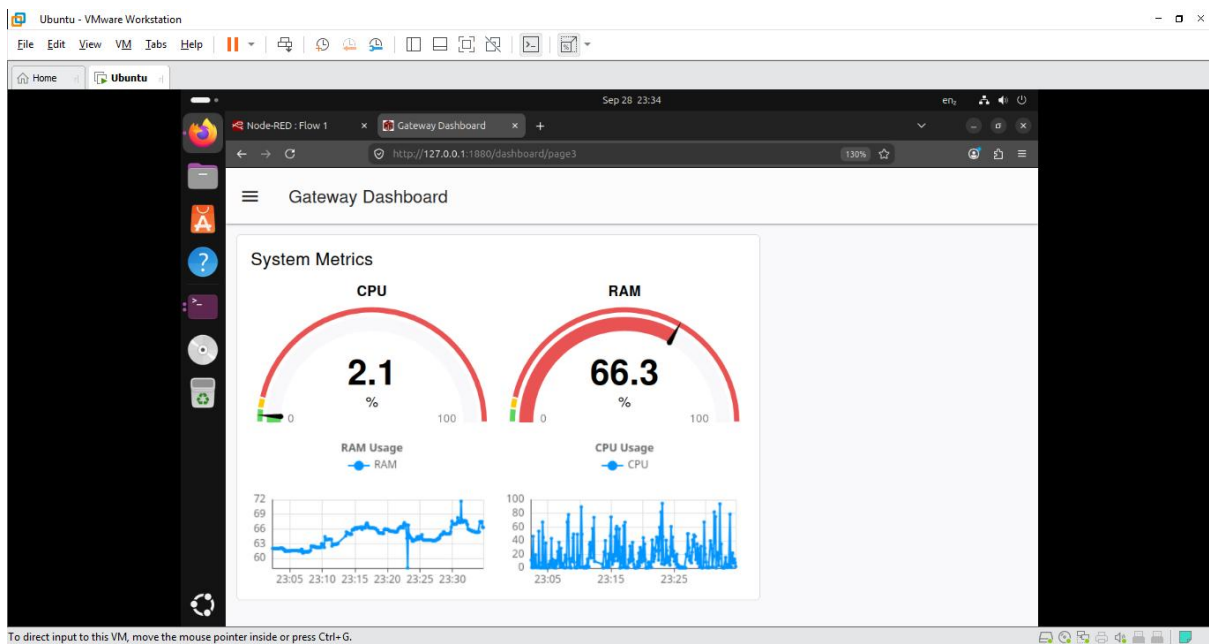
• Bài 3 (Về nhà - Nâng cao): Tích hợp Node-RED trực quan hóa dữ liệu cục bộ

```
phuongnhung@phn:~$ node-red
28 Sep 22:56:41 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:56:56 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:57:11 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:57:26 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:57:41 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:57:56 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:58:11 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:58:26 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883

phuongnhung@phn:~$ python3 /home/phuongnhung/bai4_metrics.py
File ~/home/phuongnhung/bai4_metrics.py, line 5, in <module>
  client = mqtt.client(mqtt.CallbackAPIVersion.VERSION2)
          ^^^^^
NameError: name 'mqtt' is not defined
(venv) phuongnhung@phn:~$ python3 ~/bai4_metrics.py
Traceback (most recent call last):
  File ~/home/phuongnhung/bai4_metrics.py, line 5, in <module>
    client = mqtt.client(mqtt.CallbackAPIVersion.VERSION2)
            ^^^^^
NameError: name 'mqtt' is not defined
(venv) phuongnhung@phn:~$ nano ~/bai4_metrics.py
(venv) phuongnhung@phn:~$ source ~/venv/bin/activate
python3 ~/bai4_metrics.py
Da ket noi MQTT
Da gui: {'cpu': 3.6, 'ram': 61.3}
Da gui: {'cpu': 0.5, 'ram': 61.3}
Da gui: {'cpu': 19.2, 'ram': 61.3}

28 Sep 22:58:29 - [warn] [ui-base:My Dashboard] Theme 'default' specified in page 'cie78c483573cdf3' does not exist
28 Sep 22:58:29 - [info] Started flows
28 Sep 22:58:44 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:58:59 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:59:14 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:59:29 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 22:59:44 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
28 Sep 23:00:00 - [info] [mqtt-broker:832b70bde79f7924] Connection failed to broker: mqtt://localhost:1883
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.